

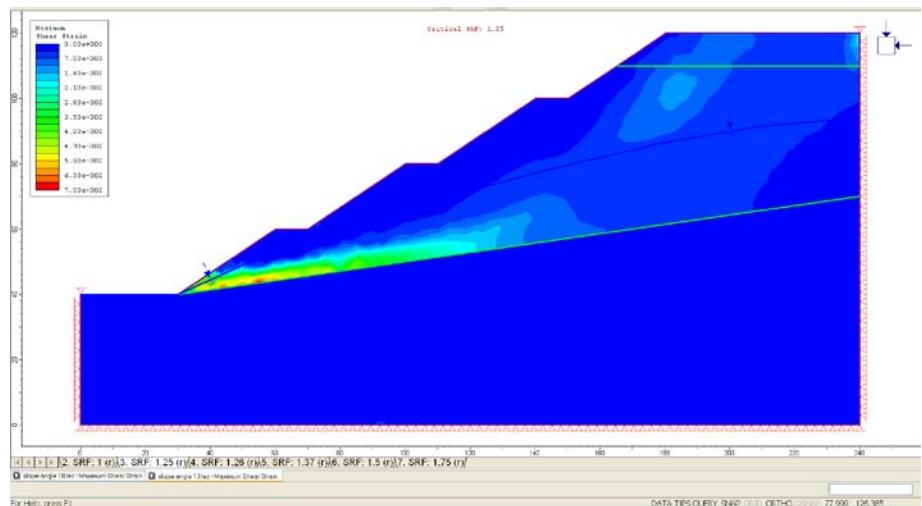
Slope Angle Optimization

In this tutorial, *Phase2* is used to examine the stability of an open pit mine. The **Change Slope Angle** editing option is used to help optimize the open pit design to make the slope as steep as possible while maintaining a suitable factor of safety.

The complete models can be found in the **Tutorial 30 Slope Angle 0.fez** to **Tutorial 30 Slope Angle 15.fez** files, which can be accessed by selecting File > Recent Folders > Tutorials Folder from the *Phase2* main menu.

Topics covered

- Importing Slide files
- Change slope angle editing option
- Batch compute
- Slope angle optimization
- Shear strength reduction



Problem

An open pit mine is excavated through weak rock. The specification of the mine design states that the factor of safety must be greater than 1.25. The goal is to create as steep a slope as possible (to minimize costs) while maintaining a factor of safety greater than 1.25. This can be done in *Phase2* by constructing a series of models with different slope angles and observing the factors of safety. By plotting the slope angle versus the factor of safety, we can obtain the optimum slope angle for the mine.

Model

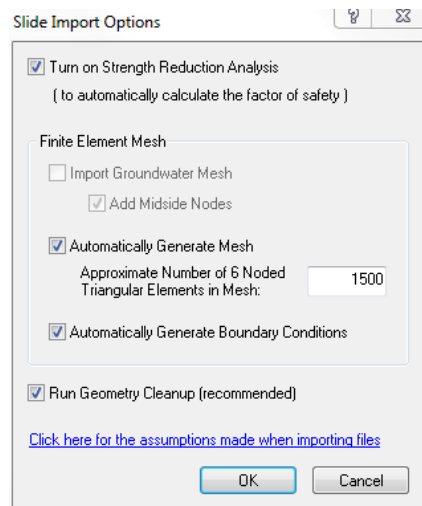
Start the *Phase2* Model program. In this tutorial we will start by importing a model created in the slope stability program *Slide* (see <http://www.rocscience.com/products/Slide.asp> for more information on *Slide*). You do not need to have *Slide* installed to import this model.

Go to **File → Import → Import Slide**. Open the *Slide* file **Tutorial 30 Slope Angle.sli** found in the Tutorials folder, which is a subfolder of the Examples folder in the *Phase2* installation directory.



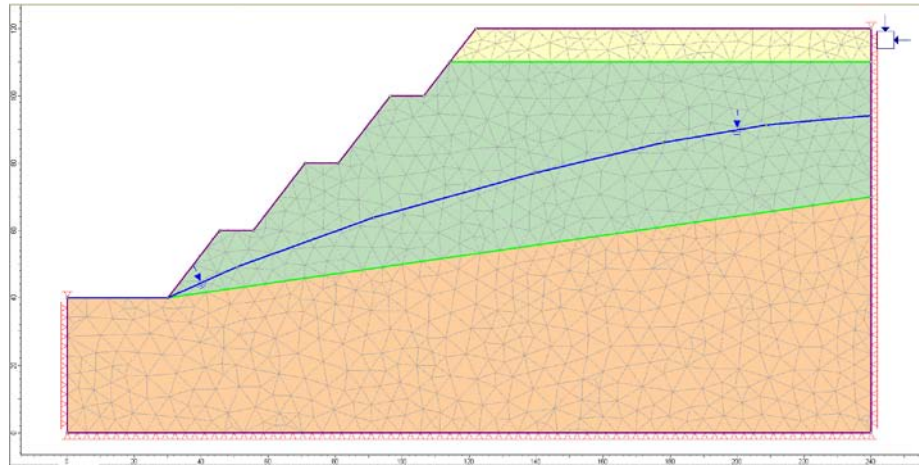
TIP: You can also import a *Slide* file by simply choosing **Open** from the **File** menu. At the bottom of the Open dialog, for **Files of type**, select **Slide File Format (*.sli)** from the drop down menu.

Once you have opened the *Slide* file, you will see the following dialog.



This allows you to set various options for the finite element analysis. We want to perform a shear strength reduction (SSR) analysis to determine the factor of safety for slope stability so leave this option on. We also want *Phase2* to automatically generate a finite element mesh and appropriate boundary conditions so leave these options on as well (*Slide* analyses do not require a finite element mesh so the mesh must be generated by *Phase2*). Click OK to accept the defaults. If a “Geometry Cleanup” dialog appears, click OK to run the Geometry Cleanup process. This is recommended when importing Slide and DXF files to ensure good mesh quality.

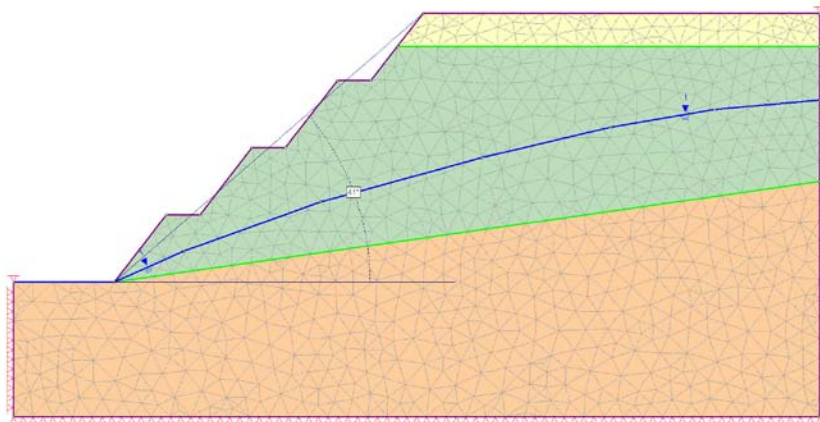
You will see a model that looks like this:



This is a model of an open pit mine in which there is a layer of sediments and a layer of weak weathered rock on top of the bedrock.



To determine the overall angle of the slope, go to **Tools → Add Tool → Dimension Angle**. Click on the crest of the slope (near 122, 120) and then click on the toe (at 30, 40). Now move the mouse to the right to draw a horizontal line. Click the mouse button to finish drawing the angle and you should see that the overall slope is approximately 41°.



Material properties

The material properties used by *Slide* are imported along with the model. However, the finite element analysis performed by *Phase2* requires some additional properties.



Go to the **Properties** menu and select **Define Materials**. Be sure you are looking at the properties for Surficial sediments. *Slide* does not require Young's modulus, Poisson's ratio or Tensile strength to perform a slope stability analysis, so default values are assigned to all materials in *Phase2*. For the Surficial sediments, change the Young's Modulus to 10000 kPa and the peak and residual Tensile Strength to 0 as shown:

The screenshot shows the 'Define Material Properties' dialog box for 'Surficial sediments'. The left pane lists materials from 'Surficial sediments' to 'Material 29'. The right pane shows the following settings:

- Name:** Surficial sediments
- Material Color:** Yellow
- Initial Element Loading:** Field Stress & Body Force
- Unit Weight: (kN/m3):** 18
- Elastic Properties:**
 - Elastic Type:** Isotropic
 - Poisson's Ratio:** 0.3
 - Young's Modulus (kPa):** 10000
 - Young's Modulus (resid) (kPa):** 20000
 - E1 (kPa):** 20000, **E2 (kPa):** 20000, **Ez (kPa):** 20000
 - v12:** 0.2, **v1z:** 0.2, **v2z:** 0.2
- Strength Parameters:**
 - Failure Criterion:** Mohr Coulomb
 - Material Type:** Plastic
 - Tensile Strength (peak) (kPa):** 0
 - Dilation Angle (deg):** 0
 - Fric. Angle (peak) (deg):** 25
 - Fric. Angle (resid) (deg):** 25
 - Cohesion (peak) (kPa):** 10
 - Cohesion (resid) (kPa):** 10
 - Tensile Strength (resid) (kPa):** 0
- Stage Properties:** ☐ (Define Factors...)
- Datum Dependent:** ☐ (Define Properties...)
- Unsaturated Shear Strength:**
 - Phi b:** 0
 - Air Entry (kPa):** 0

Buttons at the bottom: Copy To..., Statistics..., ☐ Show only properties used in model, OK, Cancel.

Now click on the tab for the Weathered Rock. Change the peak Tensile Strength to 5.

Define Material Properties

Weathered rock

Name: Weathered rock Material Color:

Initial Element Loading: Field Stress & Body Force Unit Weight: (kN/m3): 22

Elastic Properties

Elastic Type: Isotropic Poisson's Ratio: 0.3

Young's Modulus (kPa): 50000 ☐ Young's Modulus (resid) (kPa): 20000

E1 (kPa): 20000 E2 (kPa): 20000 Ez (kPa): 20000

v12: 0.2 v1z: 0.2 v2z: 0.2

Strength Parameters

Failure Criterion: Mohr Coulomb Material Type: Plastic

Tensile Strength (peak) (kPa): 5 Dilation Angle (deg): 0

Fric. Angle (peak) (deg): 28 Fric. Angle (resid) (deg): 28

Cohesion (peak) (kPa): 45 Cohesion (resid) (kPa): 45

Tensile Strength (resid) (kPa): 0

☐ Stage Properties ☐ Datum Dependent

Unsat. Shear Strength

Phi b: 0 Air Entry (kPa): 0

Copy To... Statistics... ☐ Show only properties used in model OK Cancel

Now for the Rock material, change the Young's Modulus to 200000, the Poisson's Ratio to 0.3 and the Tensile Strength to 40 kPa.

Define Material Properties

Rock

Name: Rock Material Color:

Initial Element Loading: Field Stress & Body Force Unit Weight: (kN/m3): 25

Elastic Properties

Elastic Type: Isotropic Poisson's Ratio: 0.3

Young's Modulus (kPa): 200000 ☐ Young's Modulus (resid) (kPa): 20000

E1 (kPa): 20000 E2 (kPa): 20000 Ez (kPa): 20000

v12: 0.2 v1z: 0.2 v2z: 0.2

Strength Parameters

Failure Criterion: Mohr Coulomb Material Type: Plastic

Tensile Strength (peak) (kPa): 40 Dilation Angle (deg): 0

Fric. Angle (peak) (deg): 35 Fric. Angle (resid) (deg): 35

Cohesion (peak) (kPa): 200 Cohesion (resid) (kPa): 200

Tensile Strength (resid) (kPa): 0

☐ Stage Properties ☐ Datum Dependent

Unsat. Shear Strength

Phi b: 0 Air Entry (kPa): 0

Copy To... Statistics... ☐ Show only properties used in model OK Cancel

Click OK to close the dialog.



The model definition is now complete. Save the model using the **Save As** option in the **File** menu.

Compute



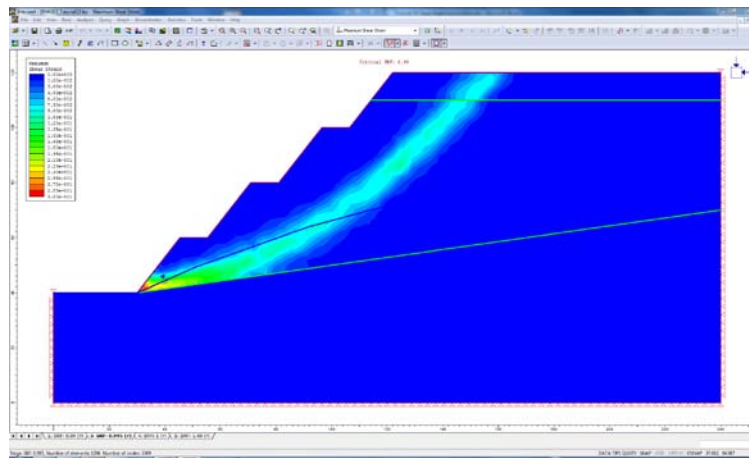
Run the model using the **Compute** option in the **Analysis** menu. Because it is performing a Shear Strength Reduction analysis, the model will take several minutes to run.



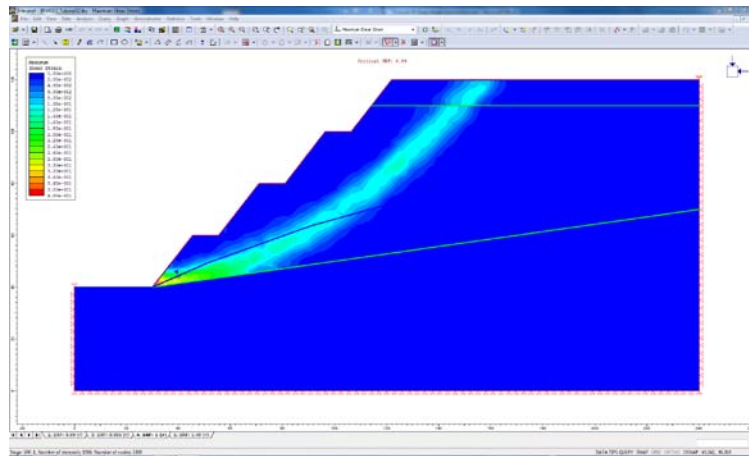
Once the model has finished computing (Compute dialog closes), select the **Interpret** option in the **Analysis** menu to view the results.

Interpret

The Interpret program starts and reads the results of the analysis. You will now see the maximum shear strain contours for the critical strength reduction factor of 0.99.



If you click the tab for SRF: 1 you get a better picture of the critical failure surface as shown.



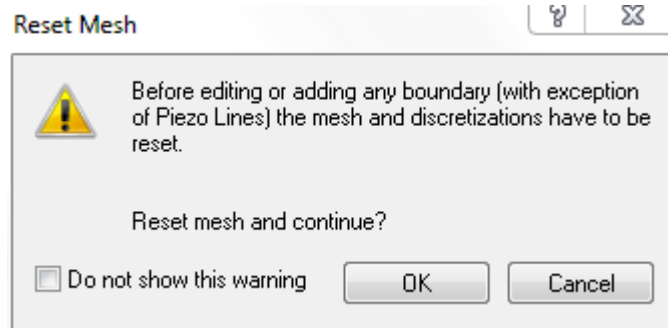
The critical SRF is equal to the factor of safety. A value of 0.99 is clearly unacceptable. We will now proceed to modify the slope angle to produce a higher factor of safety.

Changing slope angle



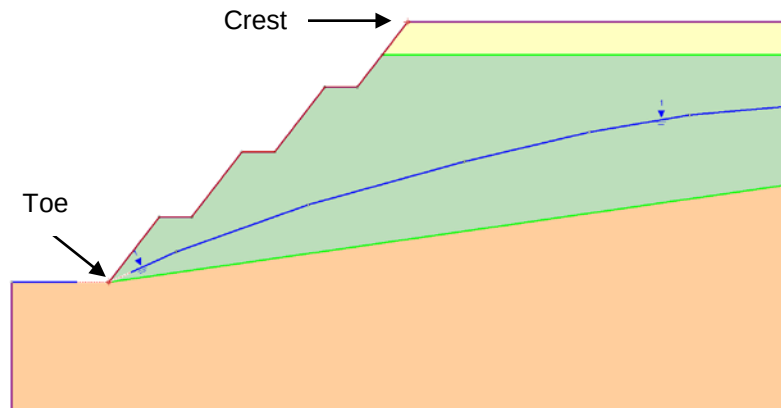
Go back to the *Phase2* Model program. First delete the Dimension Angle drawing tool by right clicking on it and selecting Delete Tool.

Go to **Boundaries** → **Edit** → **Change Slope Angle**. You will be asked if you wish to reset the mesh.



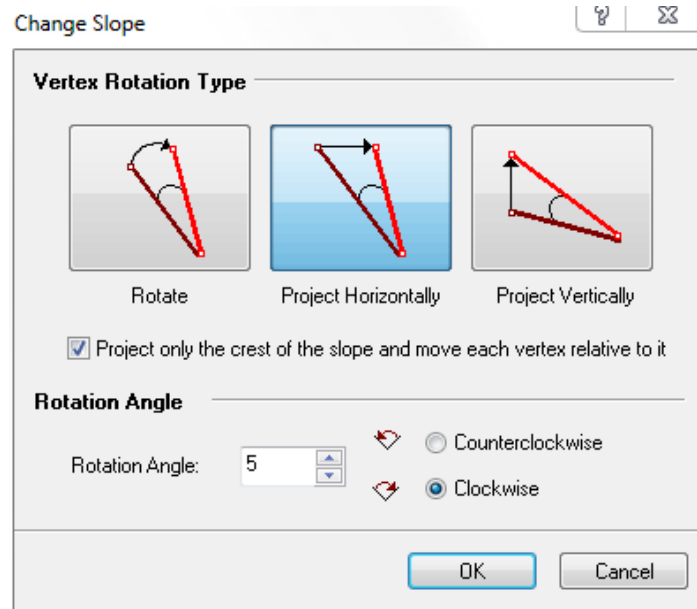
Click OK and the current mesh will be deleted.

You are now prompted to pick the starting vertex at the **toe** of the slope. Click on the point at 30 , 40. You are now asked to pick the vertex at the **crest** of the slope. Click on the point at the top of the slope (close to 122 , 120).

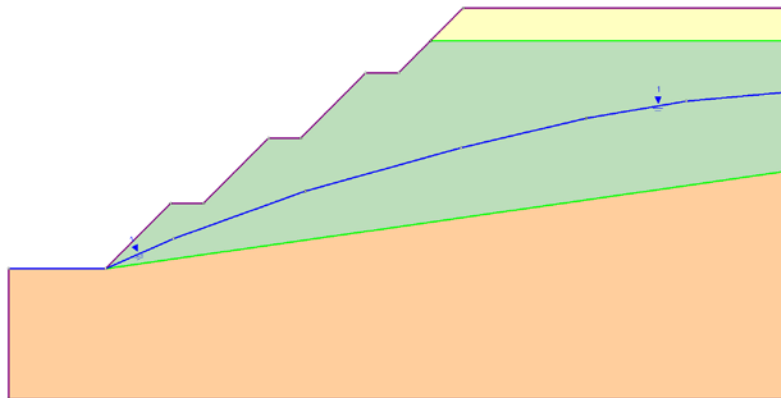


After clicking on the top point, you are presented with the **Change Slope** dialog. The default action is to **Project Horizontally**. This is what we want since we have flat benches in our slope and we do not want them to be rotated (try clicking on the option for rotate and see what happens).

We want to make the slope shallower so choose clockwise for the rotation. The default value of 5° is suitable for a first guess.



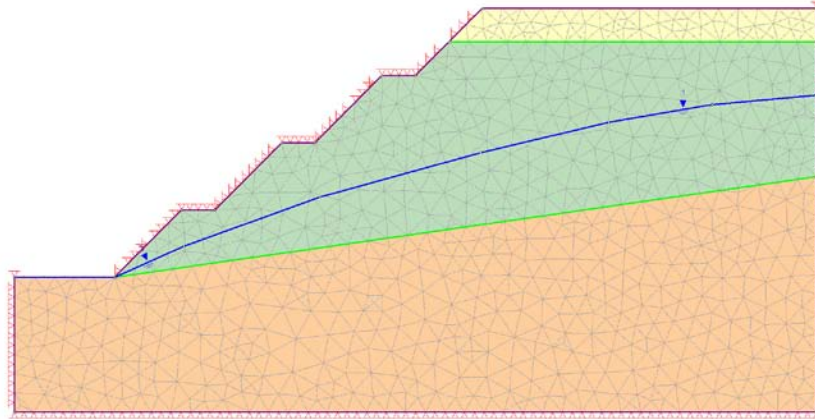
Click OK to close the dialog. You will now see that the overall slope angle is shallower.



Mesh



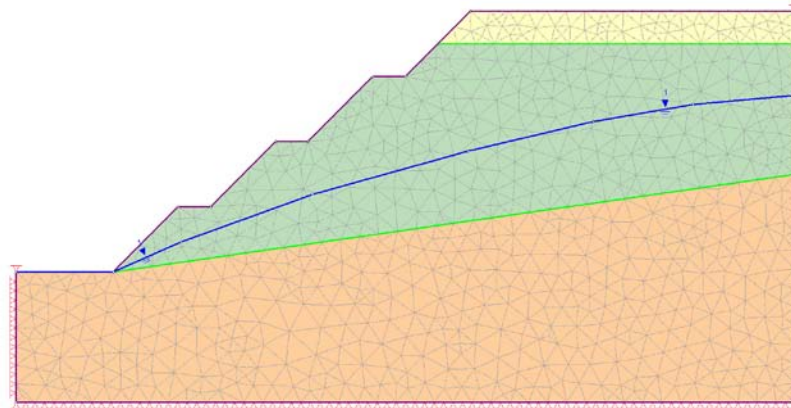
Remesh the model by selecting **Discretize and Mesh** from the **Mesh** menu. The model will look like this:



Boundary conditions



You can see that the boundary conditions on the slope face have reverted to the default (fixed in x and y direction). We must free these boundaries. Choose **Free** from the **Displacements** menu. Click on all of the segments of the slope face. Hit Enter to finish selecting segments. The model should now look like this:



Save the model using the **Save As** option in the **File** menu (give it a different name from the previous model).

Compute



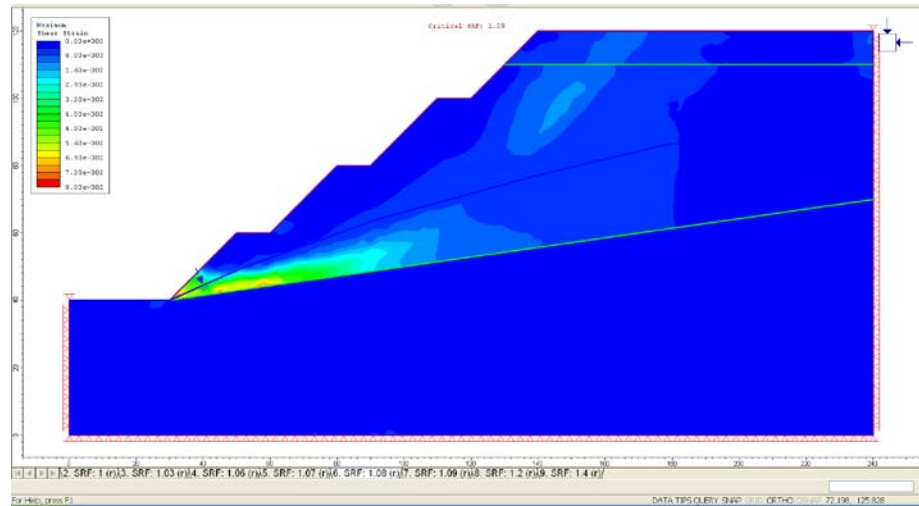
Run the model using the **Compute** option in the **Analysis** menu. Because it is performing a Shear Strength Reduction analysis, the model will take several minutes to run.



Once the model has finished computing (Compute dialog closes), select the **Interpret** option in the **Analysis** menu to view the results.

Interpret

You will see the maximum shear strain contours for the critical strength reduction factor of 1.08.



This is still less than the desired factor of safety (1.25), so we will continue to decrease the slope angle.

Slope optimization

We now want to determine what slope angle will give a factor of safety of 1.25. The best way to do this is to run a few more examples and then plot a graph of factor of safety versus slope angle. We can then interpolate to get the desired slope angle.

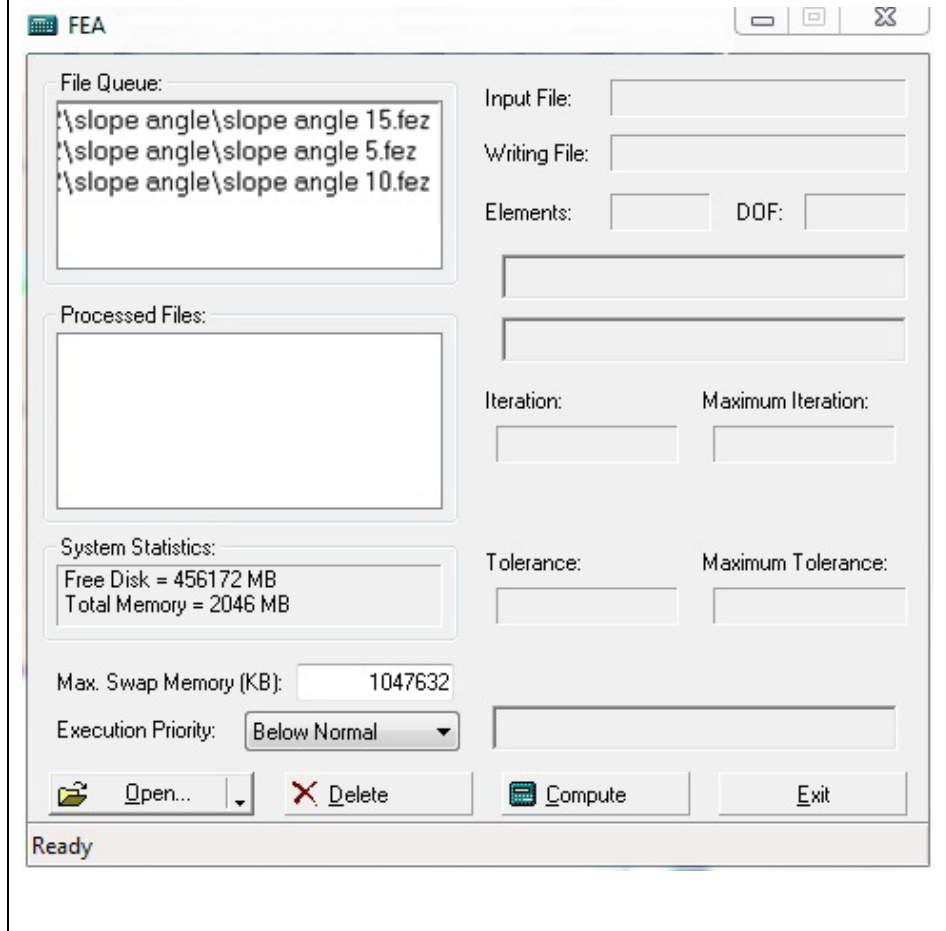


Go back to the *Phase2* Model program. Repeat the above analysis and rotate the slope by another 5° clockwise. Then repeat again rotating by another 5°. This fourth model will have a slope that has been rotated 15° from the original.

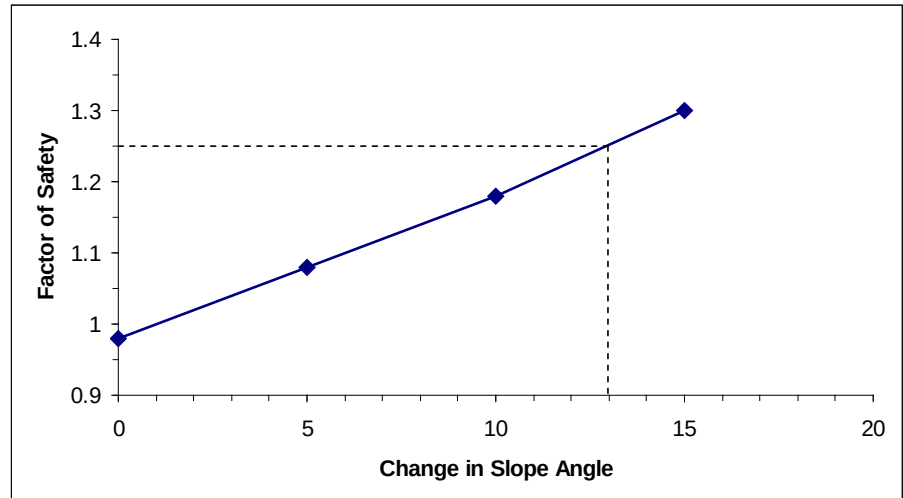
NOTE: in general, it is better to start with the **maximum** slope angle, and use the Change Slope Angle option to **decrease** the slope angle. If you do this, *Phase2* will simply crop any material boundaries at the new slope face. If you start with a shallow slope and make it steeper, *Phase2* will automatically extend any material boundaries which intersect the slope, however the results may not be as you intended, and you may have to perform additional editing to achieve the correct boundaries.

TIP: you can create multiple *Phase2* models and then run them all in batch mode. For example, after you have created your models, go the windows Start menu, and select **All Programs → Rocscience → Phase2 → Utilities → Compute**).

In the Compute dialog, you can open multiple files and then hit Compute. The program will then compute them all sequentially.



After you have finished running the models, open them in the Interpret program to determine the factors of safety. Use a spreadsheet program (e.g. Microsoft Excel) to plot Factor of Safety versus Change in Slope Angle. The plot should look like this:

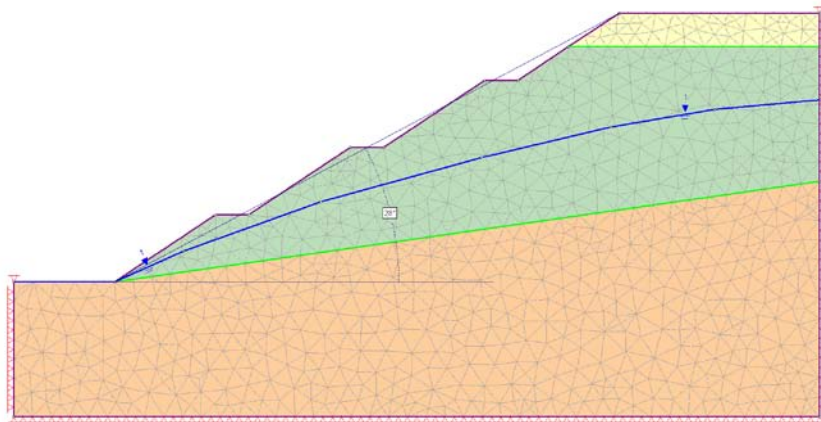


You can now interpolate to estimate the change in slope angle that will produce a factor of safety of 1.25 (shown with dashed lines in the above plot). It appears that a value of 13° should produce the desired results.

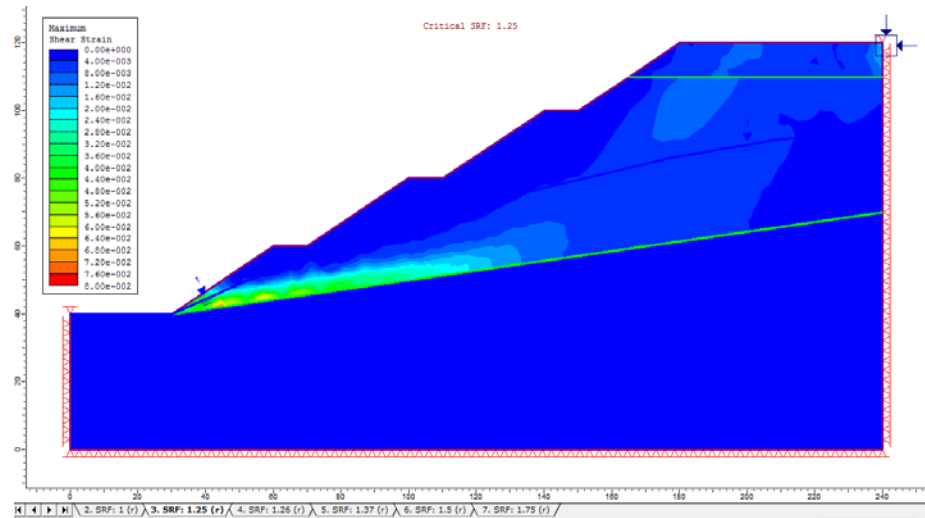
Final pit design



In the *Phase2* Model program, open the original model and rotate the slope by 13° clockwise following the steps outlined above. Measure the overall angle of the slope as you did for the first model (go to **Tools** → **Add Tool** → **Dimension Angle** and click on the crest, toe and a third point to the right of the toe). You will see that the overall angle is now 28°.



Run Compute and view the results in Interpret. You should see a factor of safety of 1.25 as expected.



This concludes the tutorial; you may now exit the *Phase2* Interpret and *Phase2* Model programs.